

CEPCC408: COMPUTER NETWORKS — ANSWER SHEET

Modules: 4, 5 & 6

Answer any 5 from Q1 | Any 3 from Q2, Q3 | Any 1 from Q4

Q.1 — Answer ANY 5 Questions (2 Marks Each)

Q1(a): Compare Coaxial Cable, Fiber Optic, and Twisted Pair Cables on their characteristics. [CO1 | BTL2]

Characteristic	Twisted Pair	Coaxial Cable	Fiber Optic
Bandwidth	Low–Med (up to 1 Gbps)	Medium (up to 10 Gbps)	Very High (100+ Tbps)
Attenuation	High	Medium	Very Low
EMI Resistance	Low (shielded better)	Medium	Excellent (immune to EMI)
Cost	Cheapest	Moderate	Expensive
Installation	Easy	Moderate	Complex
Distance	Short (~100 m/segment)	Medium (up to 500 m)	Very Long (up to 80 km)
Security	Low	Medium	High (hard to tap)
Use Case	LAN, telephone	Cable TV, older LANs	Backbone, long-haul WAN

Q1(b): Convert IP addresses from dotted-decimal to binary notation. [CO3 | BTL2]

a. 224.155.251.15

224 = 1110 0000 | 155 = 1001 1011 | 251 = 1111 1011 | 15 = 0000 1111

Binary: 11100000 . 10011011 . 11111011 . 00001111

b. 190.11.21.239

190 = 1011 1110 | 11 = 0000 1011 | 21 = 0001 0101 | 239 = 1110 1111

Binary: 10111110 . 00001011 . 00010101 . 11101111

Q1(c): Find the subnet address for the given IP and subnet mask combinations. [CO3 | BTL2]

Case 1: IP = 198.47.34.31 | Subnet Mask = 255.255.224.0

	Octet 1	Octet 2	Octet 3	Octet 4
IP Address	198 = 11000110	47 = 00101111	34 = 00100010	31 = 00011111
Subnet Mask	255 = 11111111	255 = 11111111	224 = 11100000	0 = 00000000
AND Result	11000110 = 198	00101111 = 47	00100000 = 32	00000000 = 0

Subnet Address = 198.47.32.0

Case 2: IP = 129.31.72.24 | Subnet Mask = 255.255.192.0

	Octet 1	Octet 2	Octet 3	Octet 4
IP Address	129=10000001	31=00011111	72=01001000	24=00011000
Subnet Mask	255=11111111	255=11111111	192=11000000	0=00000000
AND Result	10000001=129	00011111=31	01000000=64	00000000=0

Subnet Address = 129.31.64.0

Q1(d): List and brief any two types of query messages used by ICMP. [CO4 | BTL3]

1. Echo Request and Echo Reply (Type 8 / Type 0):

Used to test whether a destination host is reachable and responding. The sender sends an Echo Request; if alive, the destination responds with Echo Reply. This is the basis of the **ping** utility. Includes an identifier, sequence number, and optional data.

2. Timestamp Request and Timestamp Reply (Type 13 / Type 14):

Used to determine round-trip time between two hosts and to synchronize clocks. The sender includes its originate timestamp; the receiver fills in a receive timestamp and transmit timestamp on reply. Helps estimate one-way and round-trip delays.

Q1(e): Write features of BGP and define any one pros and cons. [CO4 | BTL3]

Features of BGP (Border Gateway Protocol):

- Path-vector protocol — uses AS (Autonomous System) path as the routing metric, not hop count.
- Operates over TCP port 179, ensuring reliable, ordered delivery of routing updates.
- Supports policy-based routing — routes chosen based on organisational policies, not just shortest path.
- Exchanges routing info between different Autonomous Systems (inter-domain / exterior gateway protocol).
- Uses attributes: AS_PATH, NEXT_HOP, LOCAL_PREF, MED for flexible route selection.
- Sends only incremental updates (changes), not the full table — saves bandwidth.
- Supports CIDR for flexible IP address allocation across the Internet.
- Distinguishes iBGP (within an AS) and eBGP (between different ASes).

Pros — Scalability:

BGP is the only protocol that can handle the Internet's scale, routing among hundreds of thousands of ASes. Incremental updates reduce bandwidth use, and policy-based routing gives organisations full control over traffic engineering.

Cons — Slow Convergence:

BGP converges slowly after network failures. Timer-based mechanisms (hold-down timers) mean recovery from topology changes can take minutes, causing temporary routing loops or unreachable destinations.

Q1(f): Draw UDP header format and brief any one field in detail. [CO5 | BTL3]

The UDP header is fixed at **8 bytes** (four 16-bit fields):

Source Port (16 bits)	Destination Port (16 bits)
Length (16 bits)	Checksum (16 bits)
Data (variable length)	

Field	Size	Description
Source Port	16 bits	Port number of the sending process. Optional — set to 0 if unused.
Destination Port	16 bits	Port number of the receiving process (e.g., DNS=53, HTTP=80).
Length	16 bits	Total length of UDP header + data in bytes. Minimum value = 8.
Checksum	16 bits	Error detection covering header + data + pseudo IP header. Optional in IPv4, mandatory in IPv6.

Detailed — Checksum Field:

The 16-bit checksum provides error detection for the entire UDP datagram. The sender computes a one's complement sum over a pseudo-header (source IP, destination IP, protocol number, UDP length), the UDP header, and the data. The receiver recomputes the checksum; if the result is not all 1s, the datagram is discarded. In IPv4 it is optional (field set to 0 if unused); in IPv6 it is mandatory.

Q1(g): Identify each field of the URL: <http://www.fcrit.com/Lab.html> [CO6 | BTL2]

Field	Value	Description
Protocol / Scheme	http	Specifies the application-layer protocol to use. HTTP = HyperText Transfer Protocol (default port 80). HTTPS uses port 443.
Separator	://	Standard delimiter that separates the protocol from the rest of the URL. Always three characters: colon + two forward slashes.
Host / Server Name	www.fcrit.com	"www" is the subdomain. "fcrit" is the registered second-level domain. ".com" is the Top-Level Domain (TLD) indicating a commercial organisation.
Path / Resource	/Lab.html	The specific file on the server to retrieve. "/" represents the root directory of the web server. "Lab.html" is the filename.
File Extension	.html	Indicates the file is an HTML (HyperText Markup Language) document that the browser will render as a web page.

Q.2 — Answer ANY 3 Questions (5 Marks Each)

Q2(a): 8-bit data 10101101 encoded using even parity Hamming code. Find encoded bit stream. [CO2 | BTL3]

Step 1: Determine parity bits needed

Data bits $m=8$. Need r parity bits where $2^r \geq m+r+1 \rightarrow 2^4=16 \geq 13 \checkmark \rightarrow r=4$. Parity bits at positions 1,2,4,8 (powers of 2).

Step 2: Place data bits in non-parity positions (3,5,6,7,9,10,11,12)

Position	1	2	3	4	5	6	7	8	9	10	11	12
Label	P1	P2	D1	P4	D2	D3	D4	P8	D5	D6	D7	D8
Bit	?	?	1	?	0	1	0	?	1	1	0	1

Step 3: Calculate parity bits (Even Parity — XOR of covered positions = 0)

P1 covers pos 1,3,5,7,9,11 $\rightarrow$ bits: P1,1,0,0,1,0 $\rightarrow$ XOR= $1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 = 0 \rightarrow$ **P1=0**

P2 covers pos 2,3,6,7,10,11 $\rightarrow$ bits: P2,1,1,0,1,0 $\rightarrow$ XOR= $1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 = 1 \rightarrow$ **P2=1**

P4 covers pos 4,5,6,7,12 $\rightarrow$ bits: P4,0,1,0,1 $\rightarrow$ XOR= $0 \oplus 1 \oplus 0 \oplus 1 = 0 \rightarrow$ **P4=0**

P8 covers pos 8,9,10,11,12 → bits: P8,1,1,0,1 → XOR=1⊕1⊕0⊕1=1 → **P8=1**

Position	1	2	3	4	5	6	7	8	9	10	11	12
Label	P1	P2	D1	P4	D2	D3	D4	P8	D5	D6	D7	D8
Bit	0	1	1	0	0	1	0	1	1	1	0	1

Encoded bit stream (12 bits): 0 1 1 0 0 1 0 1 1 1 0 1

Q2(b): Solve using Dijkstra's Algorithm. Graph edges: A-B=2, A-D=1, A-C=5, B-C=3, B-E=2, B-D=2, D-E=1, E-C=3, E-F=2, C-F=5. [CO4 | BTL3]

Applying Dijkstra from source A (∞ = infinity, * = permanent/finalised):

Step	Visited	A	B	C	D	E	F
Init	—	0	∞	∞	∞	∞	∞
1	A (0)	0*	2 via A	5 via A	1 via A	∞	∞
2	D (1)	0*	2 via A	5 via A	1*	2 via D	∞
3	B (2)	0*	2*	5 via A	1*	2* via D	∞
4	E (2)	0*	2*	5 via A	1*	2*	4 via E
5	C (5)	0*	2*	5*	1*	2*	4 via E
6	F (4)	0*	2*	5*	1*	2*	4*

Final Shortest Paths from A:

Destination	Shortest Cost	Path
A → B	2	A → B
A → C	5	A → D → E → C
A → D	1	A → D
A → E	2	A → D → E
A → F	4	A → D → E → F

Q2(c): Illustrate TCP 3-way handshake for connection establishment. [CO5 | BTL2]

TCP uses a **3-Way Handshake** to establish a reliable, connection-oriented session before any data transfer begins.

Step	Direction	Segment	Flags	Description
1	Client → Server	SYN	SYN=1, ACK=0, Seq=x	Client initiates connection. Sends SYN segment with a random initial sequence number x. SYN consumes one sequence number but carries no data.
2	Server → Client	SYN + ACK	SYN=1, ACK=1, Seq=y, Ack=x+1	Server accepts. Sends SYN+ACK: acknowledges client SYN (Ack=x+1) and sends its own SYN with Seq=y. Server enters SYN_RECEIVED state.
3	Client → Server	ACK	SYN=0, ACK=1, Seq=x+1, Ack=y+1	Client acknowledges server SYN (Ack=y+1). Both sides enter ESTABLISHED state. Data transfer can now begin.

Diagram:

Component	Location	Function
TELNET Client	Client side	Converts local characters into NVT (Network Virtual Terminal) format — a universal, OS-independent standard for data representation.
TCP/IP Network Stack	Both sides	Transmits NVT-formatted byte stream across the network reliably using TCP.
TELNET Server	Server side	Receives NVT-format data and converts it into the format understood by the remote OS/application.
Pseudo Terminal Driver	Server side	Invokes the appropriate shell or application on the remote machine in response to client commands.
NVT (Network Virtual Terminal)	Conceptual layer	Defines a standard for data transmission so heterogeneous OS/terminal combinations can communicate. Uses 7-bit ASCII; both sides must follow NVT rules.

Why TELNET is Used:

- Remote command-line access to computers and servers.
- Editing files and running programs on a remote machine.
- Checking open ports (e.g., telnet 192.168.0.1 443).
- Configuring network devices such as routers and switches.
- Cross-platform support — works on Windows, macOS, and Linux.

Note: TELNET sends data including passwords in plain text, making it vulnerable. **SSH (Secure Shell)** is preferred for secure remote access today.

Q.3 — Answer ANY 3 Questions (5 Marks Each)

Q3(a): Differentiate static and dynamic routing. Explain RIP protocol with example. [CO4 | BTL2]

Parameter	Static Routing	Dynamic Routing
Definition	Routes manually configured by a network administrator.	Routes automatically updated using routing protocols.
Adaptability	Cannot adapt to network topology changes.	Adapts automatically when topology changes.
Overhead	No protocol overhead; no bandwidth consumed for updates.	Consumes bandwidth to exchange routing updates.
Scalability	Suitable for small, simple networks.	Suitable for large, complex networks.
Failure Handling	No automatic rerouting on link failure.	Automatic rerouting when a link fails.
Examples	ip route command (manual)	RIP, OSPF, EIGRP, BGP

RIP (Routing Information Protocol):

Feature	Details
Type	Distance-Vector protocol (uses Bellman-Ford algorithm)
Metric	Hop count only (number of routers traversed)
Max Hops	15 hops maximum; 16 = unreachable (infinity)

Feature	Details
Update Interval	Routing table broadcast to all neighbours every 30 seconds
Versions	RIPv1 = classful RIPv2 = classless (supports CIDR and subnet masks)
Load Balancing	Supports equal-cost load balancing across multiple paths

How RIP Works — Example:

- Routers A–B–C–D are in a linear chain, each 1 hop apart.
- Initially each router knows only its directly connected networks (cost = 0 to self).
- Every 30 s, each router broadcasts its full routing table to its neighbours.
- Router A learns from B that C is 1 hop away, so A's cost to C = 1+1 = 2 hops.
- Router A learns D is 2 hops from B, so A's cost to D = 2+1 = 3 hops.
- If link B–C fails, hop count rises to 16 (infinity) and the route is invalidated.

Q3(b): Draw TCP header format and summarise each field in detail. [CO5 | BTL2]

Source Port (16 bits)			
Sequence Number (32 bits)			
Acknowledgment Number (32 bits)			
Data Offset (4b)	Resv (6b)	URG ACK PSH RST SYN FIN (6 flags)	Window Size (16 bits)
Checksum (16 bits)			
Options (variable)			
Data (variable)			

Field	Size	Description
Source Port	16 bits	Port number of the sending application process.
Destination Port	16 bits	Port number of the receiving application process.
Sequence Number	32 bits	Byte number of the first data byte in this segment. Used for ordering and reassembly.
Acknowledgment No.	32 bits	Next byte the sender of this segment expects to receive. Valid only when ACK flag is set.
Data Offset	4 bits	Length of the TCP header in 32-bit words. Tells where data starts (min=5, max=15).
Reserved	6 bits	Reserved for future use; must be zero.
Control Flags	6 bits	URG=urgent data ACK=ack valid PSH=push data RST=reset SYN=synchronise FIN=finish connection.
Window Size	16 bits	Number of bytes receiver is willing to accept. Enables flow control.
Checksum	16 bits	Error detection over TCP header + data + pseudo IP header.
Urgent Pointer	16 bits	Points to the last urgent-data byte. Valid only when URG flag is set.
Options	Variable	Optional fields: Max Segment Size (MSS), window scale, timestamps, SACK, etc.

Q3(c): What is Congestion Control? Explain Open Loop and Closed Loop methods. [CO5 | BTL2]

What is Congestion Control?

Congestion occurs when demand for network resources (bandwidth, buffers, processing) exceeds available capacity, causing packet loss and increased delay. Congestion control refers to techniques that prevent or remove this congestion to maintain network performance and throughput.

Aspect	Open Loop (Prevention)	Closed Loop (Removal)
Goal	Prevent congestion BEFORE it occurs.	Remove congestion AFTER it has occurred.
Feedback	Does NOT use feedback from the network.	Uses feedback (signals) from congested nodes.
When Applied	Policies set at connection establishment time.	Continuously monitors network; corrects on detection.
Techniques	Retransmission policy, Discarding policy, Window policy, Acknowledgment policy, Admission control.	Backpressure, Choke packet, Implicit signalling, Explicit signalling.
Example	ATM traffic shaping, Leaky/Token bucket.	TCP slow start, AIMD, RED (Random Early Detection).

Open Loop Techniques:

- **Retransmission Policy:** Careful timer design avoids premature retransmissions that add unnecessary traffic.
- **Window Policy:** Selective Repeat (resends only lost packets) is preferred over Go-Back-N (resends many) to reduce load.
- **Acknowledgment Policy:** Receiver delays ACKs (e.g., every N packets) to reduce reverse traffic.
- **Discarding Policy:** Routers selectively drop lower-priority packets to relieve pressure.

Closed Loop Techniques:

- **Backpressure:** Congested node tells its upstream neighbour to slow down; signal propagates hop-by-hop to source.
- **Choke Packet:** Congested node sends a special packet directly to the source to reduce its transmission rate.
- **Implicit Signalling:** Source infers congestion from a missing ACK (timeout) and reduces its rate. Used by TCP.
- **Explicit Signalling:** A bit in data packets is set by the congested node — either forward (to destination) or backward (to source) — to signal congestion.

Q3(d): What is domain name space and how is it organised? Justify 'Hierarchy helps in better organisation of domain names'. [CO6 | BTL2]

Domain Name Space:

A domain name space is a hierarchical, tree-structured naming system that maps human-readable host names to IP addresses. It replaces the flat name space (where all names must be globally unique and centrally managed) with a distributed, hierarchical structure.

Level	Example	Description
Root (.)	.	Apex of the DNS tree. Root servers delegate authority downward to TLD name servers.
Top-Level Domain (TLD)	.com .edu .org .in	Generic TLDs (gTLDs) such as .com for commercial, .edu for educational, .org for organisations; and country-code TLDs (ccTLDs) like .in (India), .uk (UK).
Second-Level Domain	fcrit google amazon	Registered by organisations or individuals. Must be unique within a TLD. E.g., fcrit.com, google.com. Organisation has full authority over sub-names.
Subdomain	www mail ftp	Created internally by the organisation for logical division. E.g., www.fcrit.com serves web pages; mail.fcrit.com handles email.
Hostname	server1 pc-lab3	Name of a specific device. E.g., server1.cs.fcrit.com. Resolved to a single IP address.

Justification — Hierarchy Enables Better Organisation:

- **Avoids global naming conflicts:** 'mail.google.com' and 'mail.yahoo.com' can coexist because they belong to different branches of the tree.
- **Distributes administrative control:** IANA manages the root, Verisign manages .com, FCRIT manages fcrit.com — no central bottleneck.
- **Scales to billions of devices:** A flat namespace would require a single, enormous, globally-unique registry — impossible at Internet scale.
- **Example tree:** . (root) → .com → fcrit → www.fcrit.com | mail.fcrit.com | lab.fcrit.com

Q3(e): Distinguish SMTP client and server. Explain the key role of SMTP in networking. [CO6 | BTL2]

Aspect	SMTP Client	SMTP Server
Role	Initiates the email-sending process and transmits the message.	Receives messages from clients or other servers, stores or relays them.
Port Used	Connects using outgoing port 25 (inter-server) or 587 (authenticated submission).	Listens on port 25 for incoming SMTP connections.
Actions	Issues commands: HELO/EHLO, MAIL FROM, RCPT TO, DATA, QUIT.	Responds with SMTP status codes: 250 OK, 354 start input, 550 mailbox not found, etc.
Examples	Gmail app, Outlook, Thunderbird acting as MUA (Mail User Agent).	smtp.gmail.com, Postfix, Sendmail acting as MTA (Mail Transfer Agent).
Connection Role	Actively opens TCP connection to port 25 of the destination server.	Passively waits; accepts incoming TCP connections.
Storage	Does not store messages; pushes them to the server.	Queues messages and forwards to the next MTA or hands to MDA (Mail Delivery Agent).

Key Role of SMTP in Networking:

- **Email Transmission Standard:** SMTP is the universal protocol for sending emails over the Internet. Every email client and server uses it.
- **Store-and-Forward Model:** SMTP servers queue messages and forward them hop-by-hop until they reach the recipient's server, tolerating temporary failures.
- **DNS Integration:** SMTP queries DNS MX (Mail Exchanger) records to locate the correct destination mail server for a given domain.

- **Command-Response Protocol:** Human-readable commands (MAIL FROM, RCPT TO, DATA) make SMTP easy to debug and extend.
- **Push vs Pull:** SMTP is a PUSH protocol — it sends mail from client to server. Retrieval uses POP3 or IMAP.
- **Example flow:** User composes email → SMTP client connects to smtp.gmail.com (port 587) → Gmail MTA uses DNS MX lookup → forwards to recipient's MTA → stored → retrieved via IMAP.

Q.4 — Answer ANY 1 Question (10 Marks Each)

Q4(a): ISP block 170.90.0.0/16 — distribute to Group-1: 300×128 addr, Group-2: 400×16 addr, Group-3: 300×4 addr. Find remaining addresses. [CO3 | BTL3]

ISP Block: 170.90.0.0/16 → Total addresses = 2^{16} = 65,536

Group-1: 300 businesses, each needing 128 addresses

$128 = 2^7$ → host bits = 7 → subnet mask = /25 (32–7). Block size = 128 per subnet.

Starting block: 170.90.0.0/25. Blocks used: 170.90.0.0/25, 170.90.0.128/25, 170.90.1.0/25 ... (300 blocks).

Total used: $300 \times 128 = 38,400$ addresses.

Group-2: 400 small businesses, each needing 16 addresses

$16 = 2^4$ → host bits = 4 → subnet mask = /28. Block size = 16 per subnet.

Starting after Group-1 (approx 170.90.150.0). Total used: $400 \times 16 = 6,400$ addresses.

Group-3: 300 households, each needing 4 addresses

$4 = 2^2$ → host bits = 2 → subnet mask = /30. Block size = 4 per subnet.

Starting after Group-2 (approx 170.90.175.0). Total used: $300 \times 4 = 1,200$ addresses.

Group	Customers	Addresses Each	Subnet Mask	Addresses Used
Group-1 (Medium Biz)	300	128	/25	38,400
Group-2 (Small Biz)	400	16	/28	6,400
Group-3 (Households)	300	4	/30	1,200
TOTAL USED	1,000	—	—	46,000
ISP Total (given)	—	—	/16	65,536
REMAINING AVAILABLE	—	—	—	$65,536 - 46,000 = 19,536$

Answer: 19,536 addresses are still available after all allocations.

Q4(b): Distance Vector Routing — find Router E's updated routing table. Vectors from C, D, F, H with given costs. [CO4 | BTL3]

Given — distance vectors received at Router E:

Neighbour	Distance Vector Received (A,B,C,D,E,F,G,H)	E's link cost to neighbour
C	(2, 1, 0, 2, 3, 5, 4, 3)	3
D	(6, 4, 3, 0, 1, 3, 2, 4)	2
F	(5, 2, 3, 3, 1, 0, 5, 7)	5

Neighbour	Distance Vector Received (A,B,C,D,E,F,G,H)	E's link cost to neighbour
H	(5, 2, 2, 4, 1, 3, 4, 0)	1

Formula: **Cost via X to dest = (E→X cost) + (X's cost to dest)**. Best = minimum across all neighbours.

Dest	Via C (E-C=3)	Via D (E-D=2)	Via F (E-F=5)	Via H (E-H=1)	Best Cost	Use Link
A	3+2=5	2+6=8	5+5=10	1+5=6	5	C
B	3+1=4	2+4=6	5+2=7	1+2=3	3	H
C	3+0=3	2+3=5	5+3=8	1+2=3	3	C or H
D	3+2=5	2+0=2	5+3=8	1+4=5	2	D
E	—	—	—	—	0	Self
F	3+5=8	2+3=5	5+0=5	1+3=4	4	H
G	3+4=7	2+2=4	5+5=10	1+4=5	4	D
H	3+3=6	2+4=6	5+7=12	1+0=1	1	H

Router E's Updated Routing Table:

Destination	Outgoing Link (Next Hop)	Cost
A	C	5
B	H	3
C	C (or H)	3
D	D	2
E	— (self)	0
F	H	4
G	D	4
H	H	1